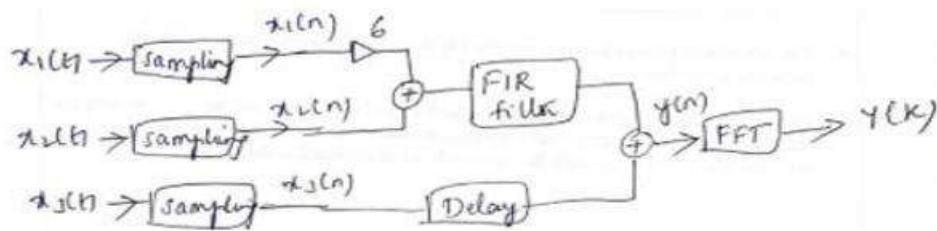


Frequency-Selective Signal Processing

Application Area: Communication and Audio Systems

FFT Analysis and FIR Low-Pass Filtering

Problem Statement:



The following data needs to be considered during the system design:

$x_1(t)$ is a sinusoid of frequency 3KHz and amplitude 3

$x_2(t)$ is a sinusoid of frequency 7KHz and amplitude 5

$x_3(t)$ is a sinusoid of frequency 9KHz and amplitude 5

The delay unit provides 1 sample delay

The FIR filter based on hamming window technique rejects frequency components above 5KHz

Introduction:

This part focuses on frequency-selective digital signal processing using sampled sinusoidal signals, FFT-based spectral analysis, and FIR filtering. Multiple frequency components are generated, combined, filtered, and analysed to demonstrate practical frequency-domain signal processing.

MATLAB Implementation:

```
clear;
clc;
close all;
f1=3000;
f2=7000;
f3=9000;
% For continuous graph visualization
t_c=0:0.0000001:0.01;
x1_c=3*sin(2*pi*f1*t_c);
x2_c=5*sin(2*pi*f2*t_c);
x3_c=5*sin(2*pi*f3*t_c);
figure
subplot(3,1,1)
plot(t_c,x1_c)
xlabel("time(sec)")
ylabel("x1(t)")
title("Continuous time domain signal x1(t)")
grid on;
subplot(3,1,2)
plot(t_c,x2_c)
```

```

xlabel("time(sec)")
ylabel("x2(t)")
title("Continuous time domain signal x2(t)")
grid on;
subplot(3,1,3)
plot(t_c,x3_c)
xlabel("time(sec)")
ylabel("x3(t)")
title("Continuous time domain signal x3(t)")
grid on;
%Sampling
fs=150000; %20khz is enough
ts=1/fs;
t=0:ts:1/1000-1/fs;
x1=3*sin(2*pi*f1*t);
x2=5*sin(2*pi*f2*t);
x3=5*sin(2*pi*f3*t);
n=0:length(t)-1;
figure
subplot(3,1,1)
stem(n,x1)
xlabel("n")
ylabel("x1[n]")
title("Sampled signal x1[n]")
grid on;
subplot(3,1,2)
stem(n,x2)
xlabel("n")
ylabel("x2[n]")
title("Sampled signal x2[n]")
grid on;
subplot(3,1,3)
stem(n,x3)
xlabel("n")
ylabel("x3[n]")
title("Sampled signal x3[n]")
grid on;
%before filtering
x11=6*x1;
x4=x11+x2;
X4=fft(x4,length(x4));

```

```

figure
N1=length(x4);
k1=0:N1-1;
subplot(2,1,1)
stem(k1,x4)
xlabel("n")
ylabel("x4[n]")
title("x4[n]=6*x1[n]+x2[n]")
subplot(2,1,2)
stem(fs*k1/N1,abs(X4))
xlabel("Frequencies(Hertz)")
ylabel("|X4(k)|")
title("Magnitude respnse of x4[k](before filtering)")
%Filter design
N=100;
hamm=hamming(N+1);
fc=5000;
b=fir1(N,2*fc/fs,"low",hamm);
figure
freqz(b,1,1024)
[h,l]=impz(b,1);
figure
stem(l,h)
xlabel("n")
ylabel("h[n]")
title("Impulse response of the filter h[n](hamming window)")
grid on;
%Filtering
x_f=conv(x4,h);
N2=length(x_f);
X_f=fft(x_f,N2);
k2=0:N2-1;
figure
subplot(2,1,1)
stem(k2,x_f)
xlabel("n")
ylabel("x_f[n]")
title("Filtered signal x_f")
grid on;
subplot(2,1,2)

```

```

stem(fs*k2/N2,abs(X_f))
xlabel("Frequencies (hertz)")
ylabel("|X_f(k)|")
title("Magnitude response after filtering of x4[n]")
grid on;
%delay for x3
x31=[0,x3(1:end)];
N3=length(x31);
if N2>N3
    x31=[x31, zeros(1,N2-N3)];
else
    x_f=[x_f, zeros(1,N3-N2)];
end
n1=0:length(x31)-1;
figure
subplot(2,1,1)
stem(n,x3)
xlabel("n")
ylabel("x3[n]")
title("Sampled signal x3[n]")
grid on;
subplot(2,1,2)
stem(n1,x31)
xlabel("n")
ylabel("x31[n]")
title("Delayed signal and zero padded signal(to add with xf[n]) x31[n]")
grid on;
%y[n] and Y(k)
y=x_f+x31;
N4=length(y);
k4=0:N4-1;
Y=fft(y,N4);
figure
subplot(2,1,1)
stem(n1,y)
xlabel("n")
ylabel("y[n]")
title("y[n]=x31[n]+x_f[n]")
grid on;
subplot(2,1,2)

```

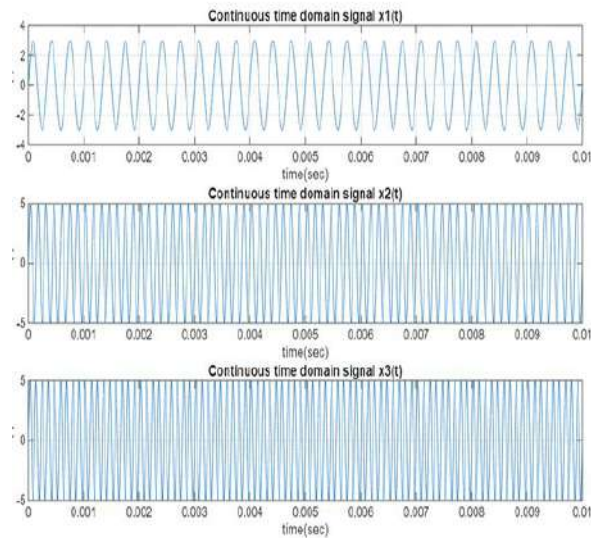
```

stem(fs*k4/N4,abs(Y))
xlabel("Frequencies(hertz)")
ylabel("|Y(k)|")
title("Magnitude spectrum of Y(k)")
grid on;

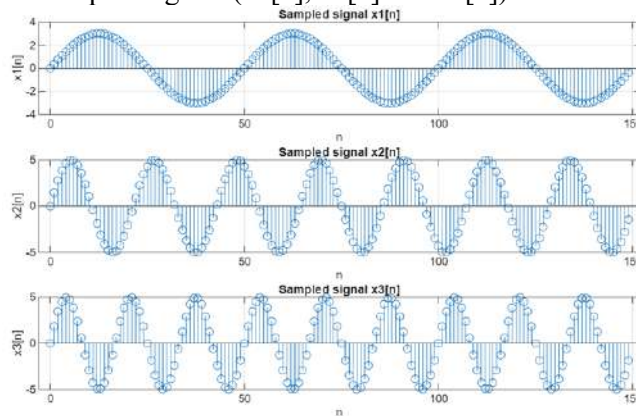
```

Results:

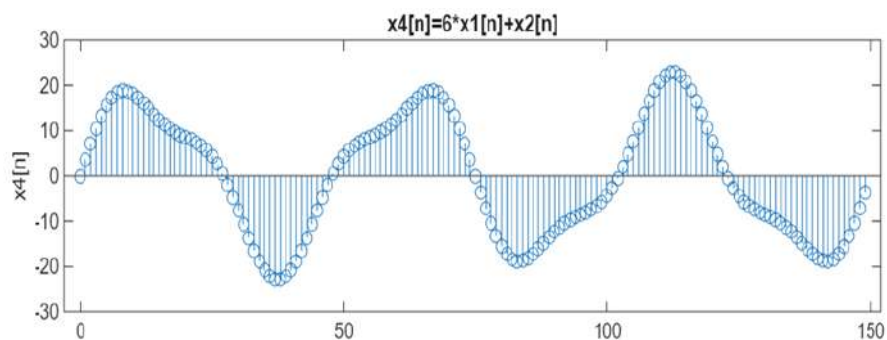
All Continuous time domain signals ($x_1(t)$, $x_2(t)$ and $x_3(t)$)

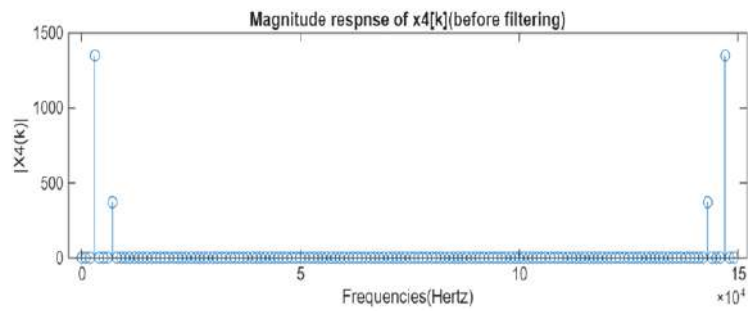


Sampled signals ($x_1[n]$, $x_2[n]$ and $x_3[n]$) $f_s=150\text{kHz}$

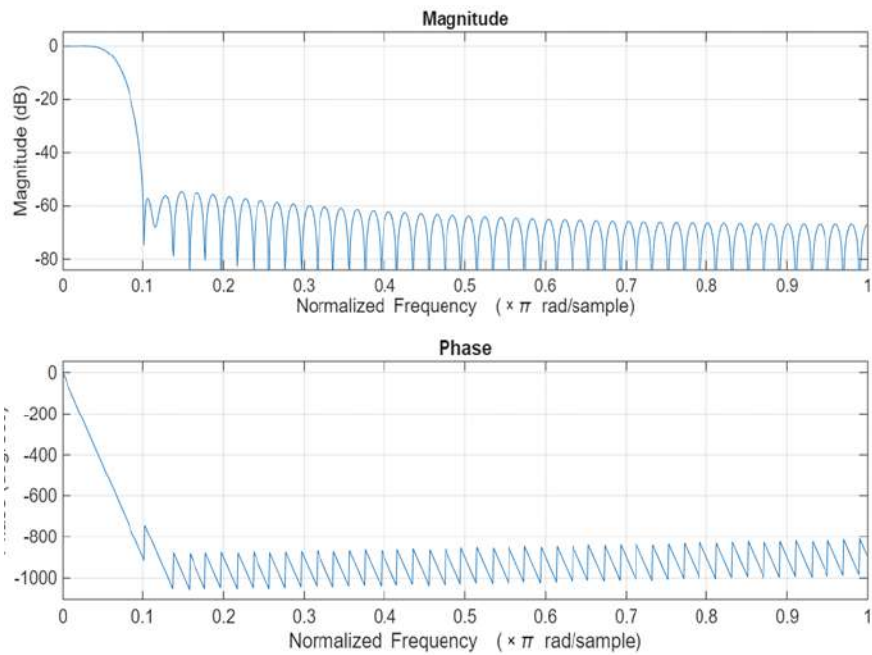


$x_4[n] = 6 \cdot x_1[n] + x_2[n]$ and magnitude response of $X_4(k)$

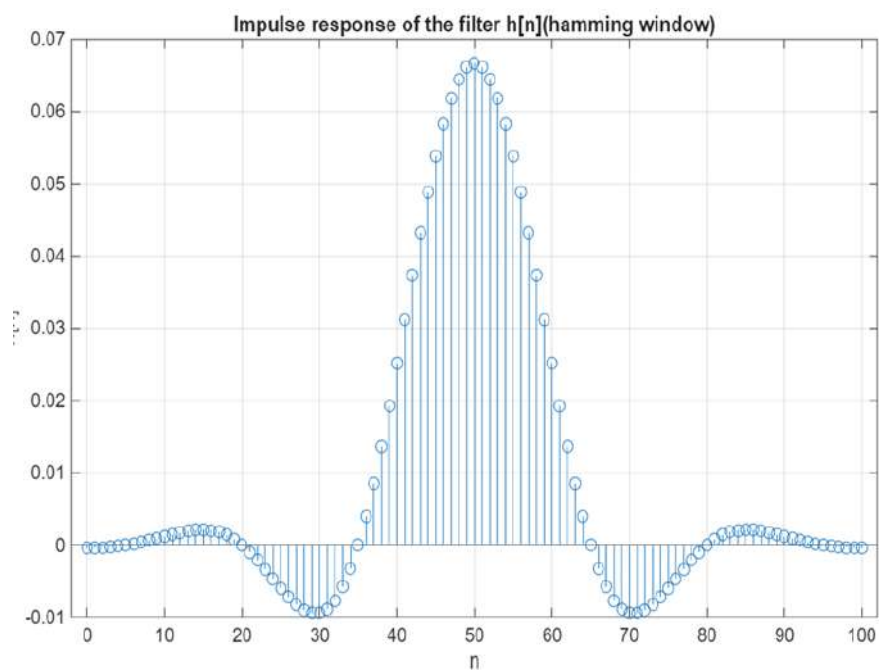




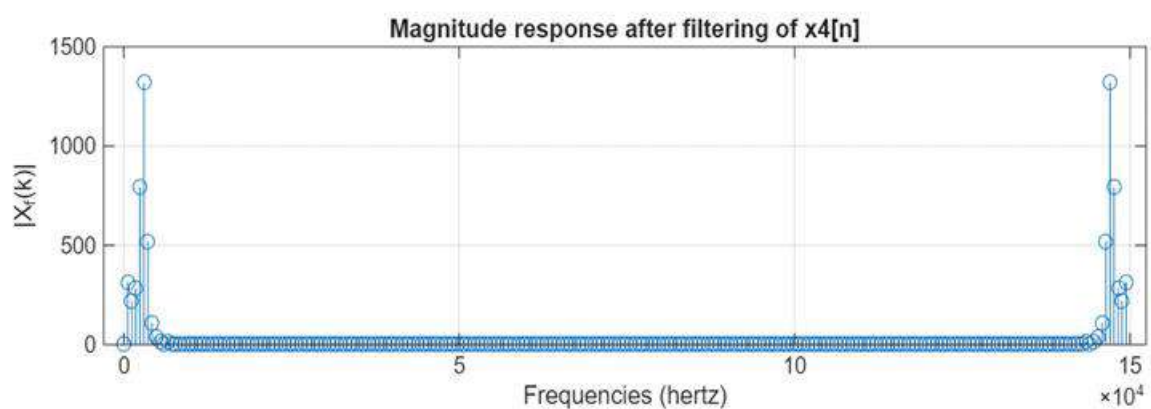
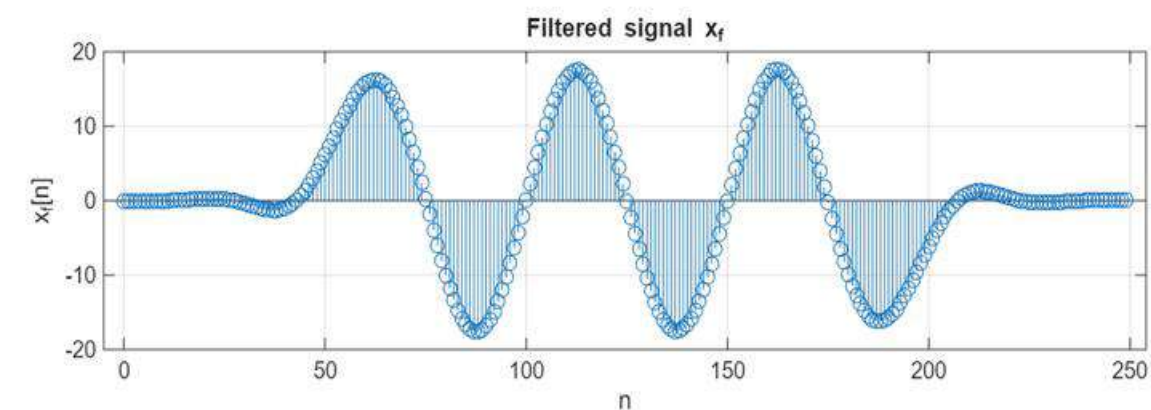
Magnitude and phase response of impulse response of the filter



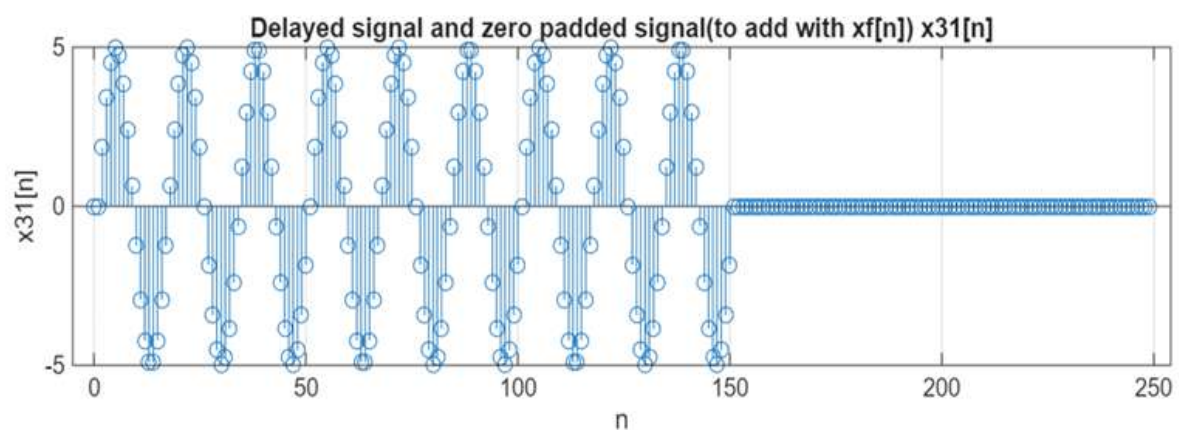
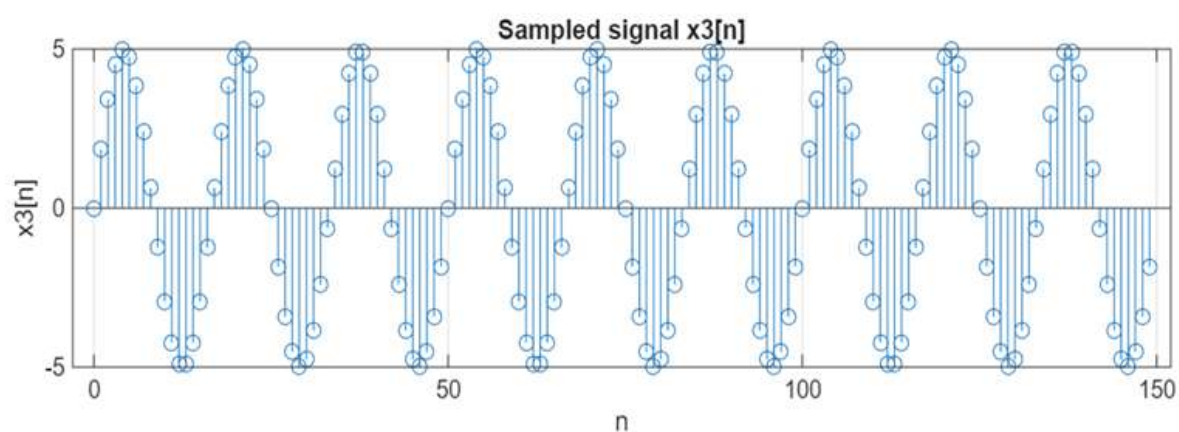
Impulse response of the filter ($h[n]$ - hamming window)



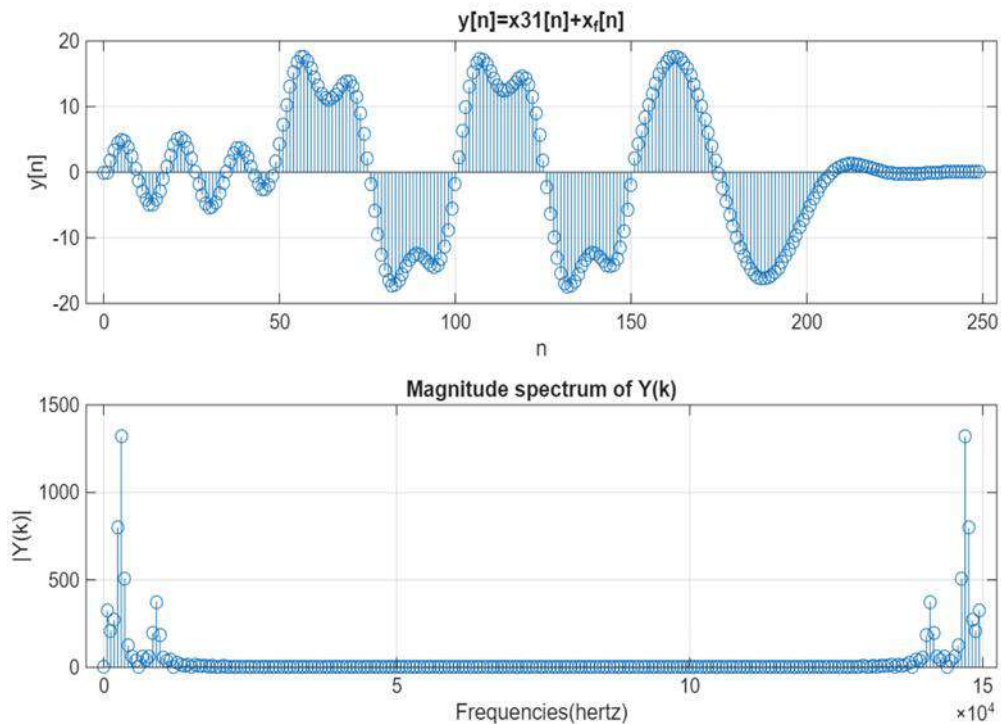
Filtered Signal $x_f[n] = \text{conv}(x_4, h)$ and magnitude response of $x_f[k]$



One sample delayed and zero padded signal $x_{31}[n]$



$Y[n] = x_{31}[n] + x_f[n]$ and magnitude response of $Y(k)$



Observations and Inferences:

- $x_1(t)$, $x_2(t)$ and $x_3(t)$ are continuous time domain signals with 3K, 5K and 9K as fundamental frequencies of each signal.
- Then Nyquist sampling rate that can be followed for the entire design is $2 \cdot \max(f_1, f_2, f_3)$ i.e., $2 \cdot \max(3K, 7K, 9K) = 18K$ Hertz, but in the design 150K Hertz as the sampling frequency.
- By perform sampling with above sampling rate we'll get $x_1[n]$, $x_2[n]$ and $x_3[n]$ from the respective signals.
- Then a new signal $x_4[n]$ is constructed as follows:

$$x_4[n] = 6 \cdot x_1[n] + x[n]$$
- In the magnitude spectrum of $X_4(k)$, we can observe 4 distinct peaks and due to conjugate symmetry property (real time signals), each frequency component duplicated twice, So it contains 2 frequency components at 3K and 7K, with 3K as the predominant one, as expected.

- To meet the required specification: To reject frequencies above 5K Hertz . Low pass filter is designed using “fir1” MATLAB’s inbuilt function.
- Order of the filter is selected in a way such that the transition band is low (Since practical implementation isn’t required)

- Then Normalized cutoff frequency is specified by the formula,

$$f_norm = 2*f_cutoff/f_sampling$$

- Hamming window of length $N+1$ is formulated using MATLAB inbuilt function `hamming(N+1)` where N is the order of the filter.

For a window length of N samples (index $n = 0, 1, 2, \dots, N - 1$):

$$w[n] = 0.54 - 0.46 \cos \left(\frac{2\pi n}{N - 1} \right)$$

- By using this snippet:

```
b=fir1(N,2*fc/fs,"low",hamming);
```

Coefficients of numerator of the filter are extracted, since FIR filter transfer function is always an all zero polynomial and all N poles of filter lie at the origin.

By using the function `freqz` , magnitude and phase response of the filter’s impulse response has been plotted.

By using the function `impz` , inverse z transform of b has done such that $h[n]$ is extracted.

- Then filtering process is done by convolving the signal $x_4[n]$ with impulse response $h[n]$, a new signal is generated (x_f). By visualizing the magnitude response of $X_f(k)$ we can observe that 7k hertz is filtered, and some unwanted frequencies with minimal amplitude are added, it is due to the truncation of the signal(hamming).
- After that $x_3[n]$ is delayed by one sample and sufficient zeros are padded to it. (to add with $x_f[n]$)
- Then $y[n] = x_f[n] + x_{31}[n]$.
- Both $y[n]$ and magnitude response of $Y(k)$ has been plotted, it can be observed that both 3k and 9k frequency components are observed, but 7k component has been filtered in the middle stage.

Work space

Workspace			
Name	Value	Size	Class
 b	<i>1×101 double</i>	1×101	double
 f1	3000	1×1	double
 f2	7000	1×1	double
 f3	9000	1×1	double
 fc	5000	1×1	double
 fs	150000	1×1	double
 h	<i>101×1 double</i>	101×1	double
 hamm	<i>101×1 double</i>	101×1	double
 k1	<i>1×150 double</i>	1×150	double
 k2	<i>1×250 double</i>	1×250	double
 k4	<i>1×250 double</i>	1×250	double
 l	<i>101×1 double</i>	101×1	double
 n	<i>1×150 double</i>	1×150	double
 n1	<i>1×250 double</i>	1×250	double
 N	100	1×1	double
 N1	150	1×1	double
 N2	250	1×1	double
 N3	151	1×1	double
 N4	250	1×1	double
 t	<i>1×150 double</i>	1×150	double
 t_c	<i>1×100001 d...</i>	1×100001	double
 ts	6.6667e-06	1×1	double
 x1	<i>1×150 double</i>	1×150	double
 x11	<i>1×150 double</i>	1×150	double
 x1_c	<i>1×100001 d...</i>	1×100001	double
 x2	<i>1×150 double</i>	1×150	double
 x2_c	<i>1×100001 d...</i>	1×100001	double
 x3	<i>1×150 double</i>	1×150	double
 x31	<i>1×250 double</i>	1×250	double
 x3_c	<i>1×100001 d...</i>	1×100001	double
 x4	<i>1×150 double</i>	1×150	double
 x_f	<i>1×250 double</i>	1×250	double
 X4	<i>1×150 comp...</i>	1×150	complex double
 X_f	<i>1×250 comp...</i>	1×250	complex double
 y	<i>1×250 double</i>	1×250	double
 Y	<i>1×250 comp...</i>	1×250	complex double